Cognitive Architecture of Speedrun: A BrainLift for the GRE Mathematics Subject Test

Owners

Amy Lin

Purpose

The purpose of this BrainLift is to map the learning science and cognitive architecture required to build "Speedrun," an innovative Anki-forked mobile and desktop application optimized for the GRE Mathematics Subject Test [1]. This document explores how practicing non-routine, high-difficulty math competition problems (such as the American Mathematics Competitions) leverages desirable difficulties [2], productive failure [3], and asymmetric cognitive transfer [4] to optimize real-test speed, conceptual flexibility, and anxiety regulation.

In Scope

- **Strategy Discrimination**: Promoting Creative Mathematically-founded Reasoning (CMR) over rote imitative reasoning [5].
- **Asymmetric Transfer**: Training on complex, low-stability tasks to generalize downward to simpler, standard test items [4].

Out of Scope

- **Passive Explanations**: Bypassing productive struggle via passive AI-generated solutions [5].

Experts

- **Dr. Robert Bjork & Dr. Elizabeth Ligon Bjork (UCLA)**: Pioneers of the "desirable difficulties" framework, distinguishing immediate performance from long-term learning [2].
- **Dr. Doug Rohrer (University of South Florida)**: Lead clinical researcher on interleaved practice and strategy discrimination in mathematics [7].
- **Dr. Manu Kapur (ETH Zurich)**: Originator of the "Productive Failure" instructional

model showing that struggling with complex problems prior to instruction yields superior conceptual understanding [3].
- **George Pólya**: Groundbreaking mathematician who codified universal mathematical problem-solving heuristics and active student inquiry [6].

DOK 1

Learning Science and Desirable Difficulties

- **Source 2**: *Making Things Hard on Yourself, But in a Good Way: Creating Desirable Difficulties to Enhance Learning (Bjork & Bjork, 2011)* [2]
  - **DOK 1 - Facts**:
    - Performance (immediate, **context-bound** execution during study) is a highly unreliable index of actual learning (durable retention and far transfer) [2].
    - Memory representations possess two properties: storage strength (how deeply integrated a memory is) and retrieval strength (current ease of access) [2].
    - Desirable difficulties—such as spacing, interleaving, and retrieval testing—deliberately challenge retrieval strength during study to maximize long-term storage strength [2].
- **Source 3**: *The Effects of Interleaved Practice (Taylor & Rohrer, 2010)* [7]
  - **DOK 1 - Facts**:
    - Fourth-grade students completing interleaved math practice scored nearly double on delayed exams (77%) compared to those who used blocked practice (38%) [7].
    - Blocked practice acts as a cognitive "crutch" because students learn how to *use* a strategy but never practice *choosing* it based on the problem itself [7].
    - Interleaving forces strategy discrimination by ensuring that consecutive problems cannot be solved using the same mathematical procedure [7].
- **Source 4**: *Productive Failure in Learning Math (Kapur, 2014)* [3]
  - **DOK 1 - Facts**:
    - Students who engage in exploratory, challenging problem-solving *before* being taught formal mathematical procedures show significantly higher conceptual understanding and transfer capability than those taught first [3].
    - Attempting to solve problems beyond one's current knowledge triggers a "failure-based activation," prompting students to analyze structural features and prepare cognitive pathways to encode expert solutions [3].
    - Early direct instruction restricts creative thinking, leaving students locked into single imitative methods [3].
- **Source 5**: *Gaining Mathematical Understanding: The Effects of Creative Mathematical*

Reasoning and Cognitive Proficiency (Jonsson et al., 2020) [5]

- **DOK 1 - Facts**:
  - Standard math curricula rely heavily on imitative "Algorithmic Reasoning" (AR), with textbook analyses showing 79% of tasks are solvable by repeating pre-specified steps [5].
  - Practicing "Creative Mathematically-founded Reasoning" (CMR)—where students must construct original solution methods—forces an effortful struggle that significantly outperforms AR on novel transfer tests [5].
  - The superior performance of CMR is primarily driven by the cognitive benefits of "effortful struggle" (large effect size of $d = 1.34$) [5].

Cognitive Transfer Dynamics

- **Source 6**: *Hard-to-Easy Effect in Auditory Discrimination Learning and Knowledge Transfer (PNAS, 2010)* [4]
  - **DOK 1 - Facts**:
    - Training on highly difficult, complex tasks benefits performance on subsequent simple target tasks by transferring generalized procedural knowledge and information-integration rules [4].
    - Progression from hard to easy tasks yields robust, asymmetric knowledge transfer, whereas progressing from easy to hard tasks yields zero upward transfer [4].
    - High-difficulty training environments foster broad cognitive heuristics and error-filtering strategies that drastically reduce cognitive workload when transitioning to simpler tasks [4].
- **Source 7**: *Why Participate in Math Competitions? (AwesomeMath)* [6]
  - **DOK 1 - Facts**:
    - Competition mathematics trains students to approach unfamiliar problems systematically by emphasizing pattern recognition, abstraction, and out-of-the-box strategy development [6].
    - Preparing for and sitting through highly challenging, multi-hour math contests builds immense mental stamina and eliminates standardized test anxiety [6].
    - Standardized test formats (like the SAT, ACT, and Subject GRE) feel highly manageable and less stressful by comparison because competition participants are accustomed to structural uncertainty under a clock [6].

DOK 2

Desirable Difficulties and the Trap of Procedural Fluency

Traditional GRE math preparation relies on blocked, scaffolded practice sets that prioritize error-free, immediate performance. While this linear progression makes students feel confident, it is a dangerous illusion of mastery [2]. Immediate success is driven by high "retrieval strength," which is temporary and context-bound [2]. True, durable learning requires introducing "desirable difficulties" during study sessions [2]. By shuffling different question types (interleaving) and distributing review sessions across time (spacing), we force students to continually retrieve information from long-term memory when retrieval strength is low, substantially increasing long-term "storage strength" and knowledge transfer [2].

Productive Failure and the Power of Creative Reasoning

Engaging students in complex, non-routine mathematical reasoning *before* providing explicit solutions is highly effective for building structural comprehension [3]. Under the "productive failure" framework, attempting tasks that are just beyond one's current mastery activates prior knowledge and clearly exposes the boundaries of existing schemas [3]. This struggle is not a sign of failure but a powerful cognitive catalyst: it primes the brain, creating "curiosity gaps" that make subsequent explicit instruction highly memorable [3]. By forcing students to construct original mathematical strategies (Creative Mathematically-founded Reasoning) instead of repeating pre-packaged textbook steps (Algorithmic Reasoning), we ensure they build flexible, deep mathematical schemas that translate into long-term academic gains [5].

Asymmetric Transfer and Cognitive Optimization under timed Constraints

Cognitive science demonstrates that learning transfer is radically asymmetric [4]. Training on simple, routine problems offers zero upward transfer to difficult tasks, whereas training on highly complex, non-routine tasks generalizes downward instantly [4]. By practicing using moderately difficult AMC problems, students are forced to abstract generalizable problem-solving heuristics and robust, flexible schemas [6]. Consequently, when these students encounter the standard, highly predictable questions of the GRE Mathematics Subject Test, their cognitive load remains low [6, 4]. The actual GRE exam items are perceived as simplified, routine subproblems of their broader, competition-forged schemas, dramatically reducing test anxiety and optimizing speed under severe timed constraints [6, 4].

DOK 3: Insights

So far, standardized math preparation does not use interleaving, instead relying on solving sections of the same type of problem. This is a huge missed opportunity to train the critical skill of **strategy selection** [7]. It forces memorization instead of true understanding; they can blindly "plug and chug" the formula they’re currently working with [7]. Interleaved competition problems, by contrast, force the brain to constantly reload different mathematical schemas into working memory [2]. Shifting continuously between disparate topics (e.g., multivariable calculus, abstract algebra, and combinatorics) forces students to compare, contrast, and

discriminate between structural task features [7]. This active strategy discrimination is the ultimate bottleneck on standardized math tests [7].
Another issue is standardized test anxiety, which is caused when a student encounters an unfamiliar, non-routine question and immediately experiences a threat response, which activates the amygdala and depletes the working memory resources needed to solve the problem [7]. When intentionally training on hard difficulty competition problems, we force students to become comfortable with the sensation of not knowing what to do at first glance [6, 3]. Cognitive tension becomes active engagement (productive struggle) rather than an impending threat, which could cause students to freeze up during the exam [3]. This psychological reframing, combined with the hard-to-easy transfer effect, makes standard exam questions feel incredibly manageable, converting test-day panic into calm execution [6, 4].

DOK 4: Spiky Point of View

To amply prepare for the GRE math subject test, students must use extreme difficulty practice such as mid-level AMC problems. Even though they are not likely to solve it, which initially discourages students, it pays off: this forced difficulty makes the GRE exam seem like a breeze.

Where my idea comes from: GRE math subject test has exactly the same format as AMC competition math questions. It is about 3 minutes per question, rapid but deep pattern recognition, no room for mistakes. I've also seen firsthand how competition math has trained me as well as others to excel at standardized tests.
This spiky POV tends to be unpopular because of concerns over discouragement of learning and mental health impact. Initially, doing competition math problems leads to rather heavy discouragement. Out of 25 questions given on the AMC exam, students solve an average of 6 problems. But it is important to ignore this feeling because it gives a better result in the long run. I believe that maintaining a positive attitude and having grit is more than enough to combat this, and are important skills to learn and exercise throughout life. Since the GRE exam is made for prospective graduate students, they should deal with discouragement/challenges better than say middle schoolers.

Other notes:
Standard test prep uses highly scaffolded, linear, and non-interleaved practice designs. It minimizes student error but introduces the illusion of understanding. Students mistake easy recognition of textbook-format questions for durable, transferable understanding [2]. Because the actual GRE Mathematics Subject Test contains approximately 66 complex, multi-disciplinary questions to be solved under severe time constraints [6], success depends on rapid strategy selection, not just procedural execution [7].

To build authentic mathematical resilience, students must train with non-routine competition math problems (e.g., AMC-level tasks) [6]. Although competition-level problems are significantly harder and initially increase practice error rates, they act as an optimal desirable difficulty [2]. This systematic over-challenge forces the cognitive abstraction of highly flexible, robust schemas [4, 6]. Under this "hard-to-easy" transfer paradigm, standard GRE questions are perceived on test day as simplified, routine subproblems of their broader schemas, radically reducing working memory depletion and completely neutralizing test anxiety [6, 4].

Bibliography

1. Bjork, E. L., & Bjork, R. A. (2011). *Making Things Hard on Yourself, But in a Good Way: Creating Desirable Difficulties to Enhance Learning*. Psychological Bulletin. URL: https://bjorklab.psych.ucla.edu/wp-content/uploads/sites/13/2016/04/EBjork_RBjork_2011.pdf
2. Taylor, K., & Rohrer, D. (2010). *The Effects of Interleaved Practice*. Applied Cognitive Psychology. URL: https://www.retrievalpractice.org/strategies/2017/interleaving
3. Kapur, M. (2014). *Productive Failure in Learning Math*. Cognitive Science. URL: https://pubmed.ncbi.nlm.nih.gov/24628487/
4. Jonsson, B., Granberg, G., & Lithner, J. (2020). *Gaining Mathematical Understanding: The Effects of Creative Mathematical Reasoning and Cognitive Proficiency*. Frontiers in Psychology. URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC7775304/
5. PNAS. (2010). *Hard-to-Easy Effect in Auditory Discrimination Learning and Knowledge Transfer*. Proceedings of the National Academy of Sciences. URL: https://www.pnas.org/doi/10.1073/pnas.0912357107
6. AwesomeMath. *Why Participate in Math Competitions?* URL: https://www.awesomemath.org/why-participate-in-math-competitions/